

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

INDEX

Sr. No.	Description	Page No.
1.	Methodology	1
	1 Satellite SAR System with Resolution and Revisit Time	
	2 Retrieving Multi-Temporal SAR Imagery Over the Specified Period	
	3 Differential SAR Interferometry (D-InSAR) for Displacement Measurement	2
2.	Visualization	3
	1 Defining specific bridge location & parameters for satellite SAR based inspection	
	2 Calibration, georeferencing, orthorectification of SAR images (VH/VV)	4
	3 Results of SAR based bridge inspection, including visualizations	5
	3.1 VV Displacement From 23 Jan 2025 to 28 June 2025	6
	3.2 VH Displacement From 23 Jan 2025 to 28 June 2025	6
	3.3 SAR Based Analysis at Abutment 1 LHS	7
	3.4 SAR Based Analysis at Abutment 1 RHS	8
	3.9 SAR Based Analysis at Abutment 2 LHS	9
	3.10 SAR Based Analysis at Abutment 2 RHS	10
3.	Analysis	11
	1 Identify potential deformation, shifts or movement of the bridge structure	
	Conclusion	

METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
S1A_IW_SLC__1SDV_20250604T130237_20250604T130307_059497_0762D9_3AB5
S1A_IW_SLC__1SDV_20250616T130237_20250616T130306_059672_0768D1_F57D
S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan

Bridge No. – 899/1

Bridge Chainage – 938+100

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – VUP

Sar Parameters

Sensor – Sentinel-1 (C-Band)

Beam Mode – IW (Interferometric Wide Swath)

Path – 34

Frame – 519

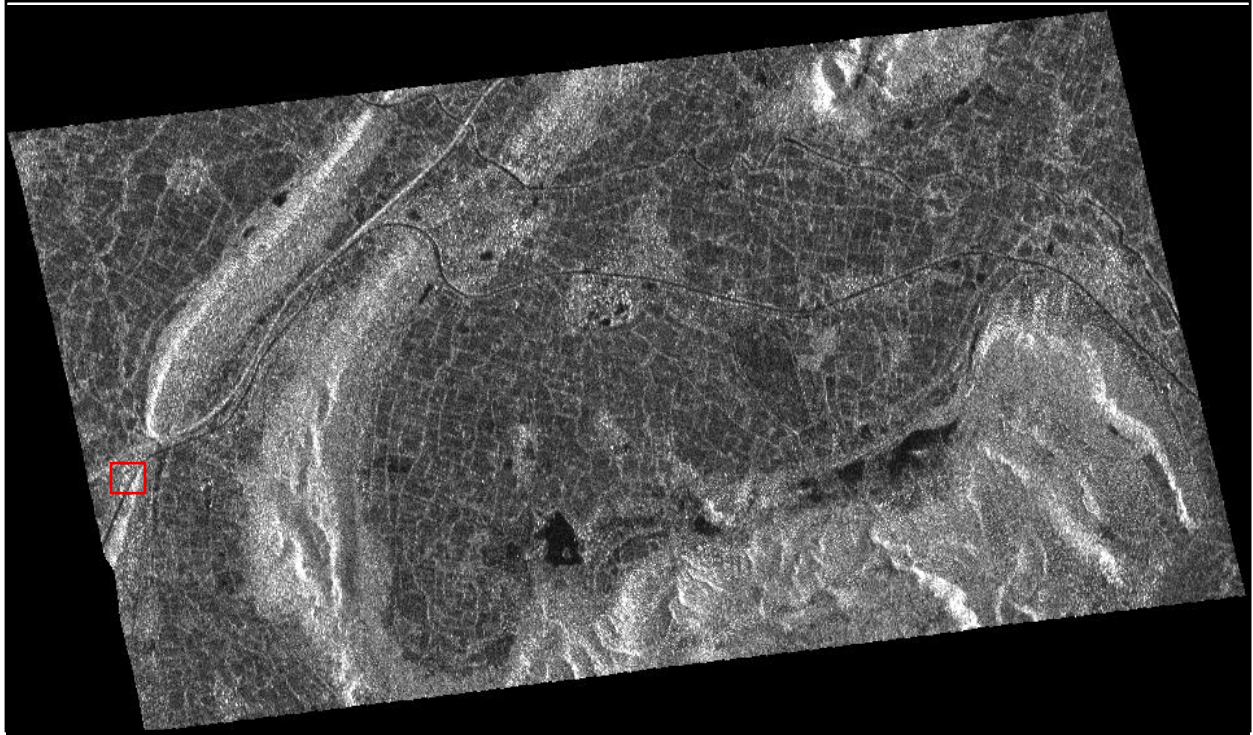
Flight Direction - DESCENDING

Polarization – VV+VH

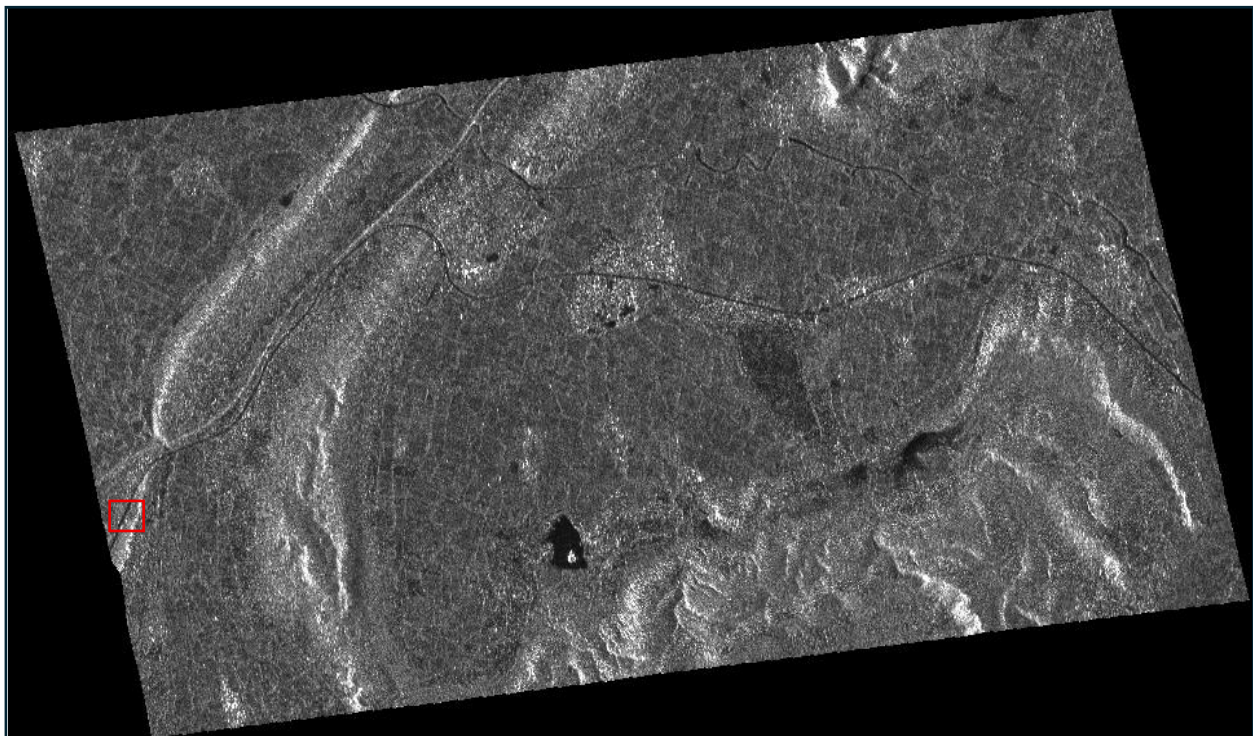
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization

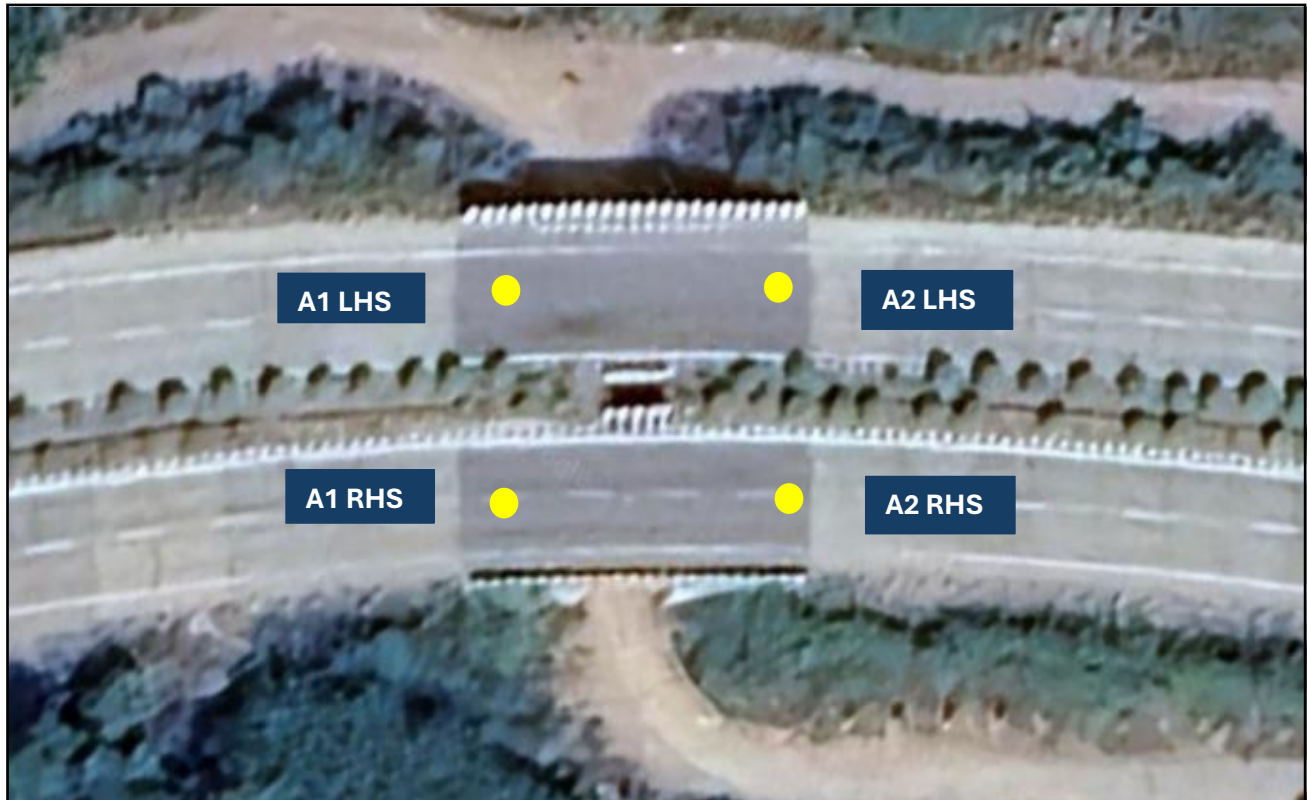


3. Results of SAR based bridge inspection, including visualizations

Bridge Components

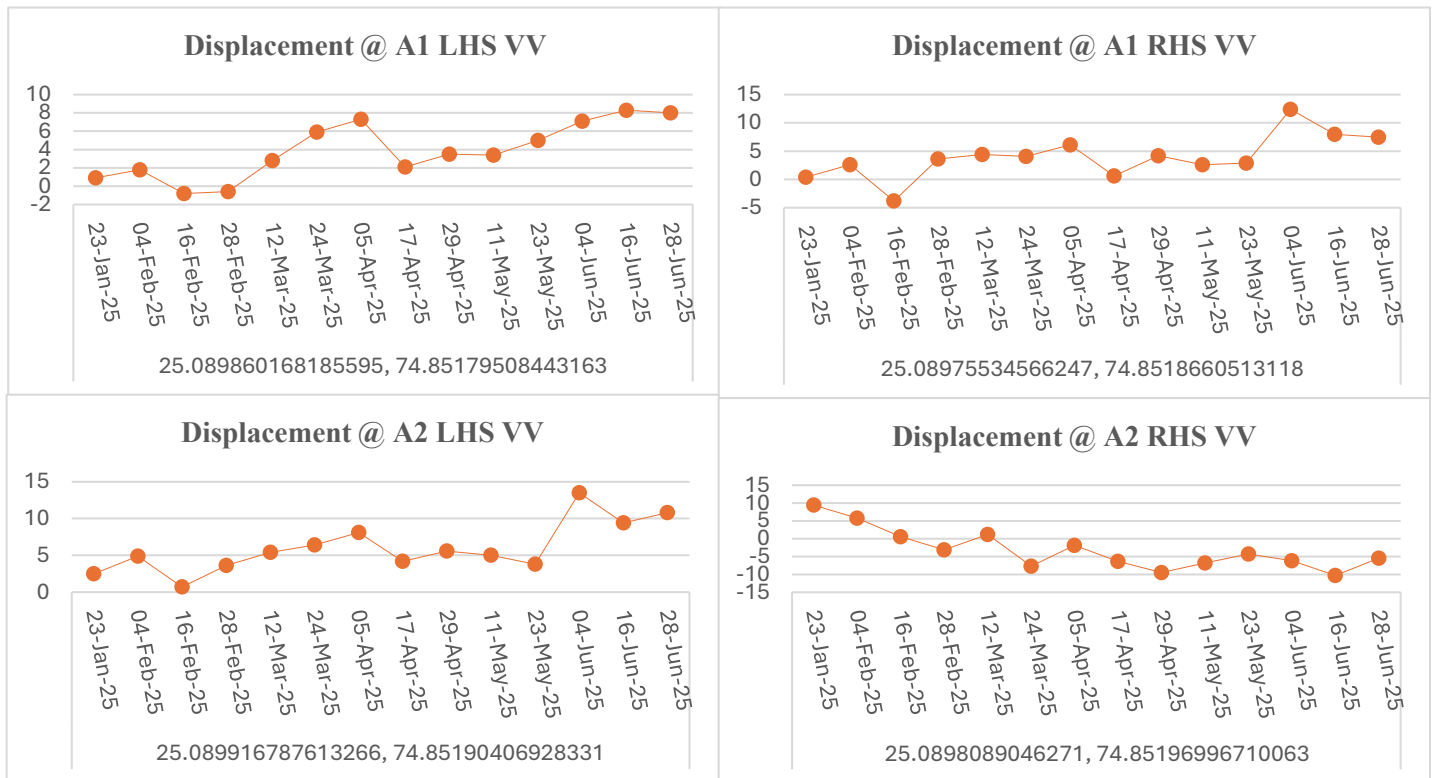
VUP- CH-938+100

Length – 23.03M

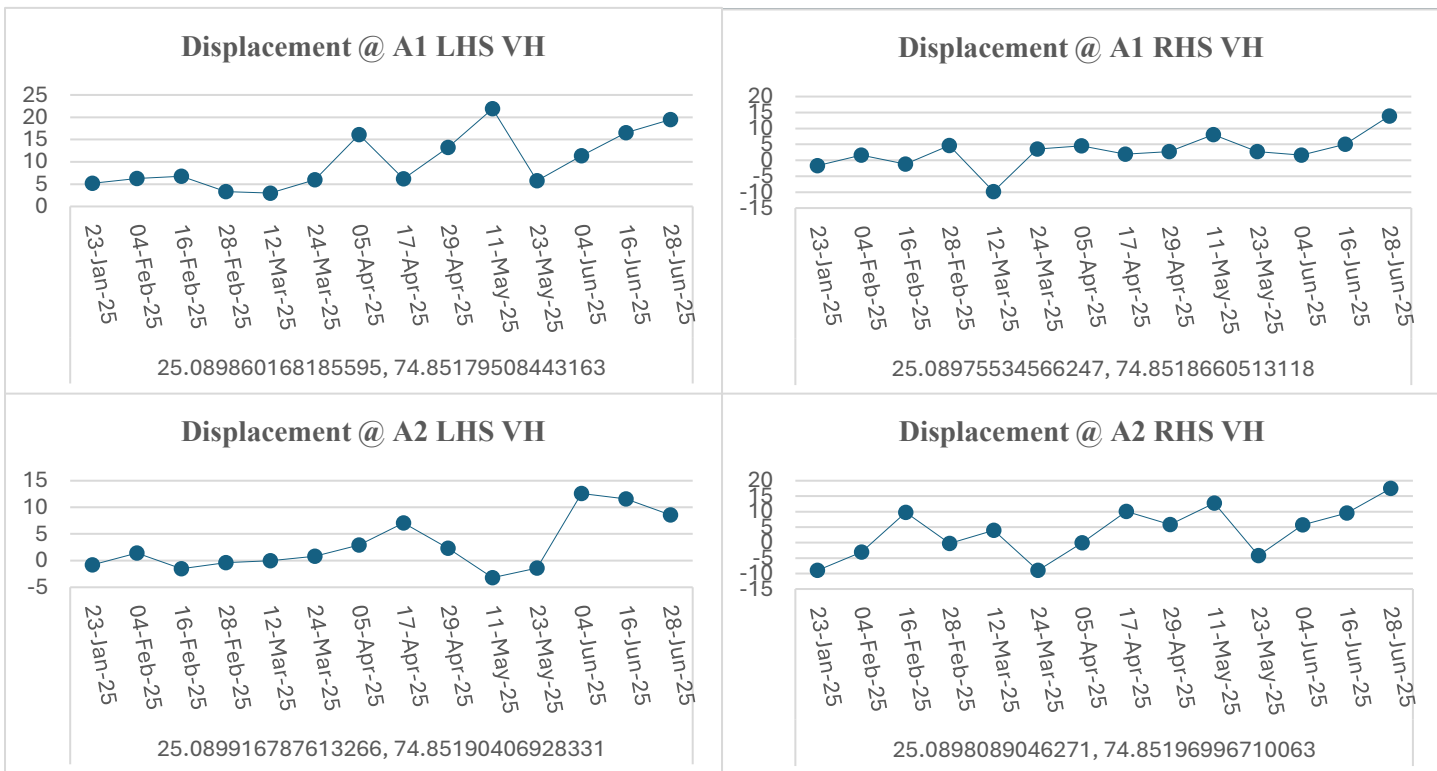


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1		Abutment 1	
A2		Abutment 2	
LHS		Left Hand Side	
RHS		Right Hand Side	

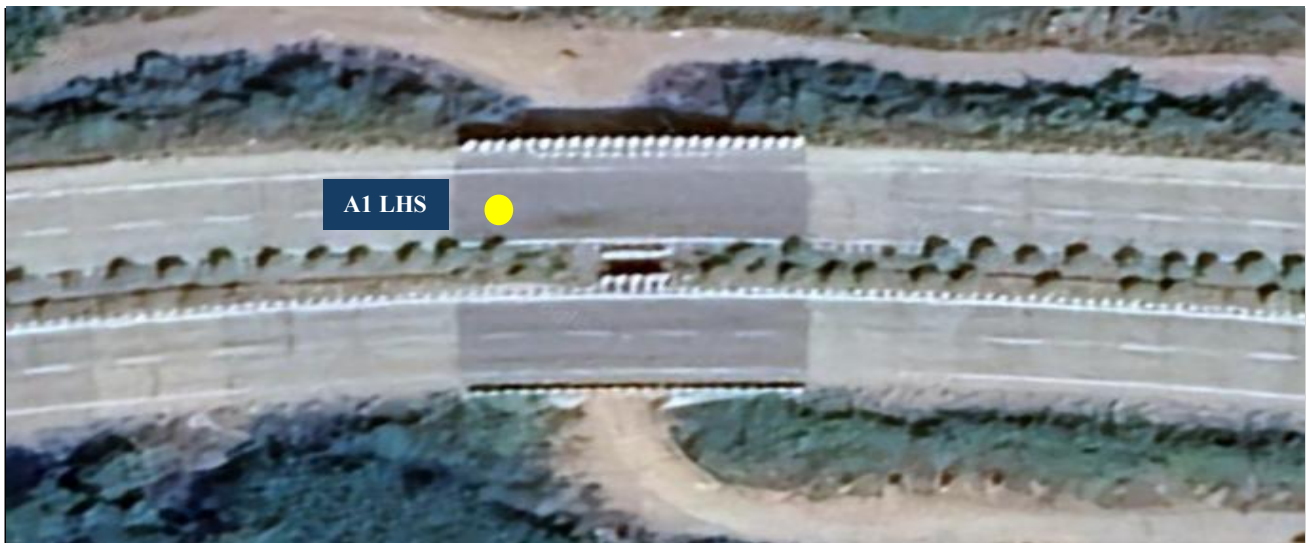
3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



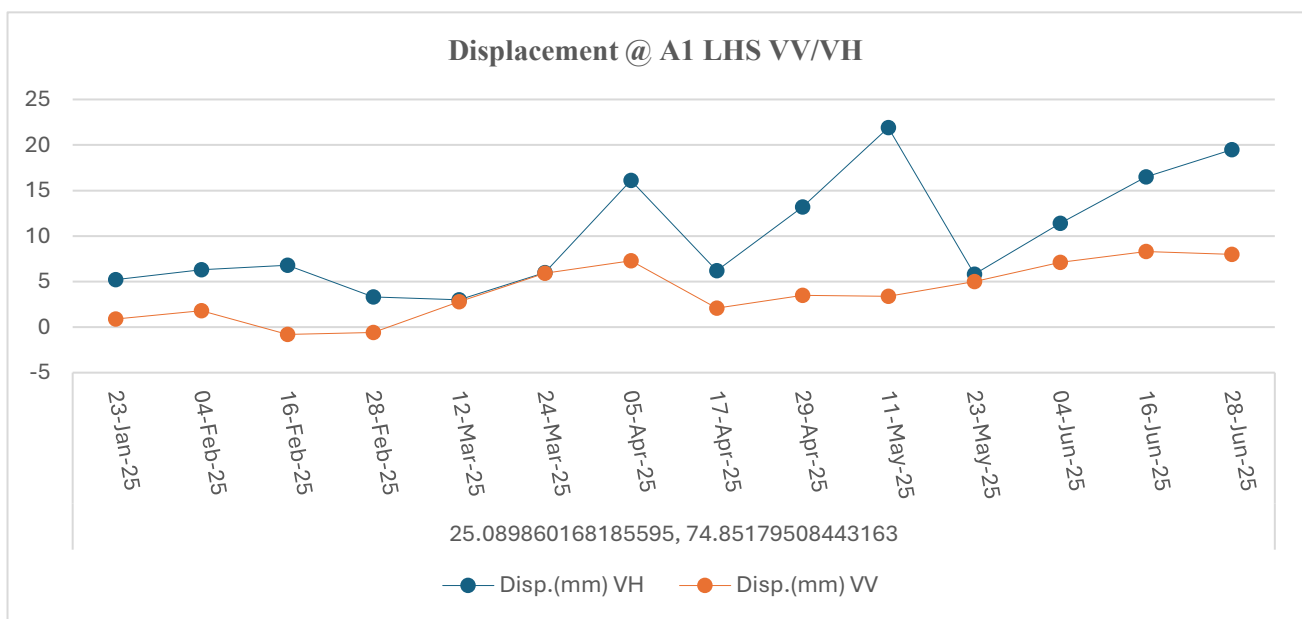
3.2 VH Displacement From 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (25.089860168185595, 74.85179508443163)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement remains comparatively moderate, varying roughly between 0 mm and +8 mm.
- **Horizontal Displacement (VH)** Horizontal displacement shows a clear increasing trend with noticeable fluctuations, ranging approximately from +3 mm to +21 mm.

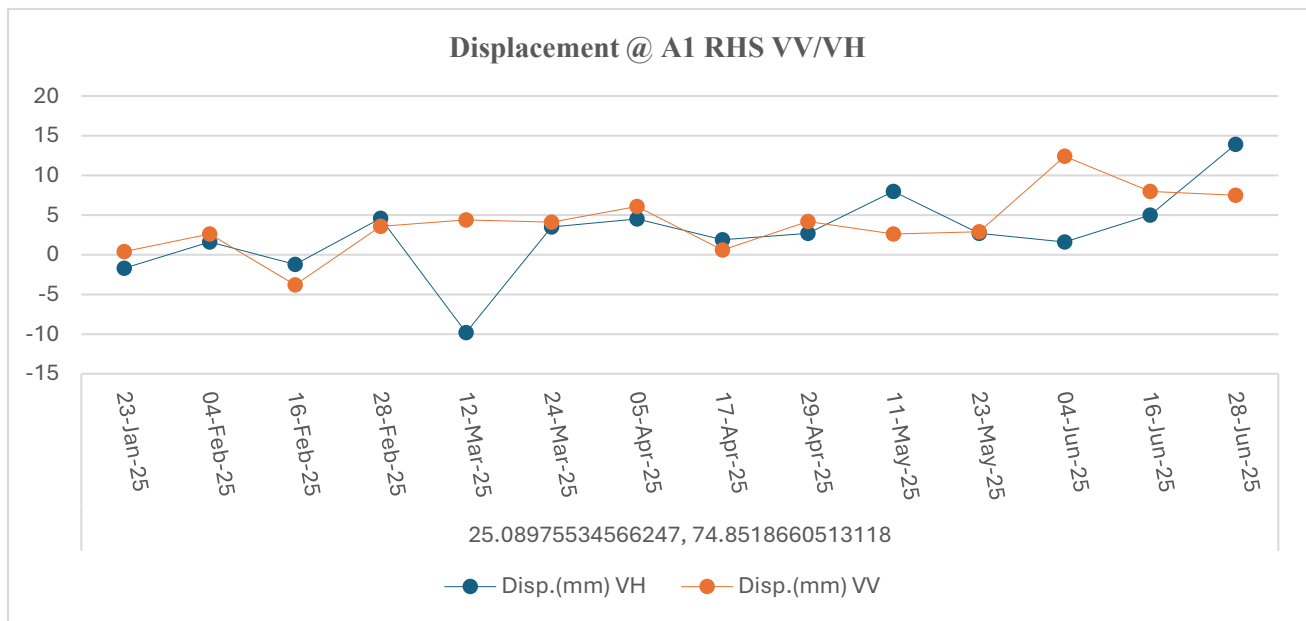
Critical Events & Anomalies:

- **February 2025:** VH shows a slight reduction (around +3 mm) while VV remains near zero, indicating a short phase of reduced lateral activity.
- **April 2025:** A sudden drop in VH with stable VV indicates partial lateral recovery or redistribution of forces.
- **May-June 2025:** Both VH and VV show a gradual increasing trend, with VH rising again to 18–19 mm and VV stabilizing around 7–8 mm.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (25.08975534566247, 74.8518660513118)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** VH exhibits moderate variability with values ranging approximately from -10 mm to +13 mm.
- **Vertical displacement (VV)** remains predominantly **positive**, varying roughly between -4 mm and +12 mm.

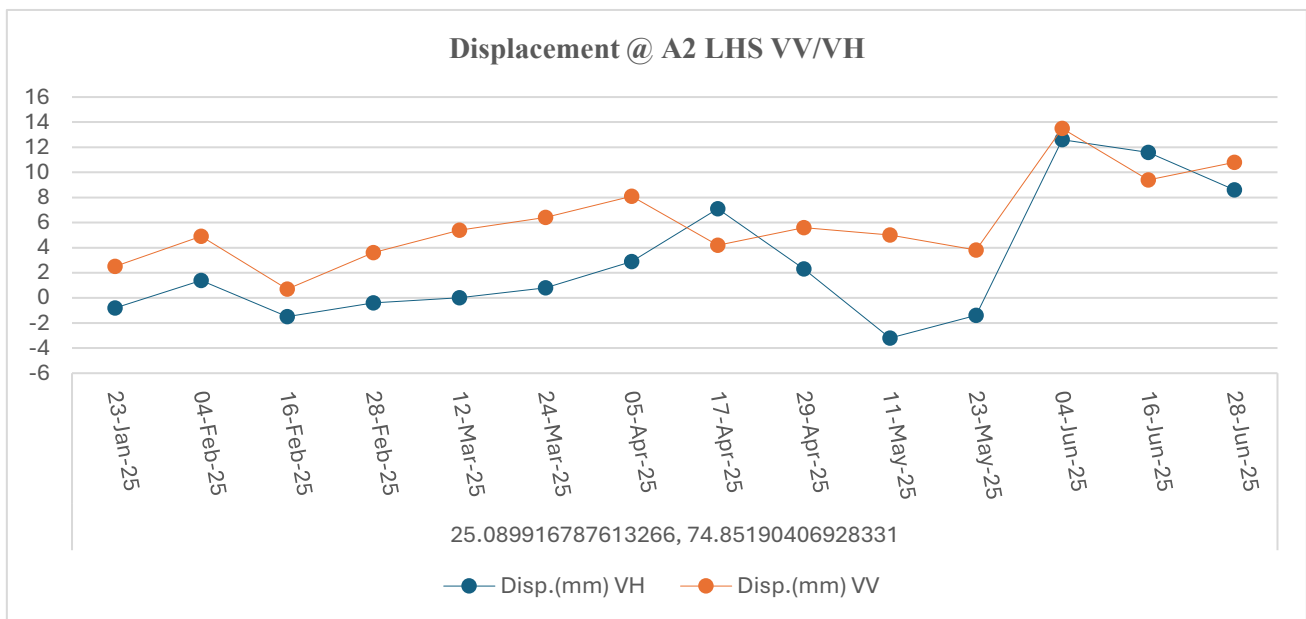
Critical Events & Anomalies:

- **February 2025:** Both VH and VV remain low and close to zero, reflecting relatively stable structural behaviour during this period.
- **March-April 2025:** VH recovers to positive values with VV remaining consistently positive, suggesting redistribution of horizontal forces with stable vertical response.
- **June 2025:** VH rises to its maximum with VV stabilizing around ~7–8 mm, pointing to renewed lateral movement without abnormal vertical distress.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (25.089916787613266, 74.85190406928331)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** shows low to moderate variability ranging approximately from -4 mm to $+13$ mm.
- **Vertical Displacement (VV)** is consistently positive throughout the monitoring period, varying roughly between $+1$ mm and $+13$ mm.

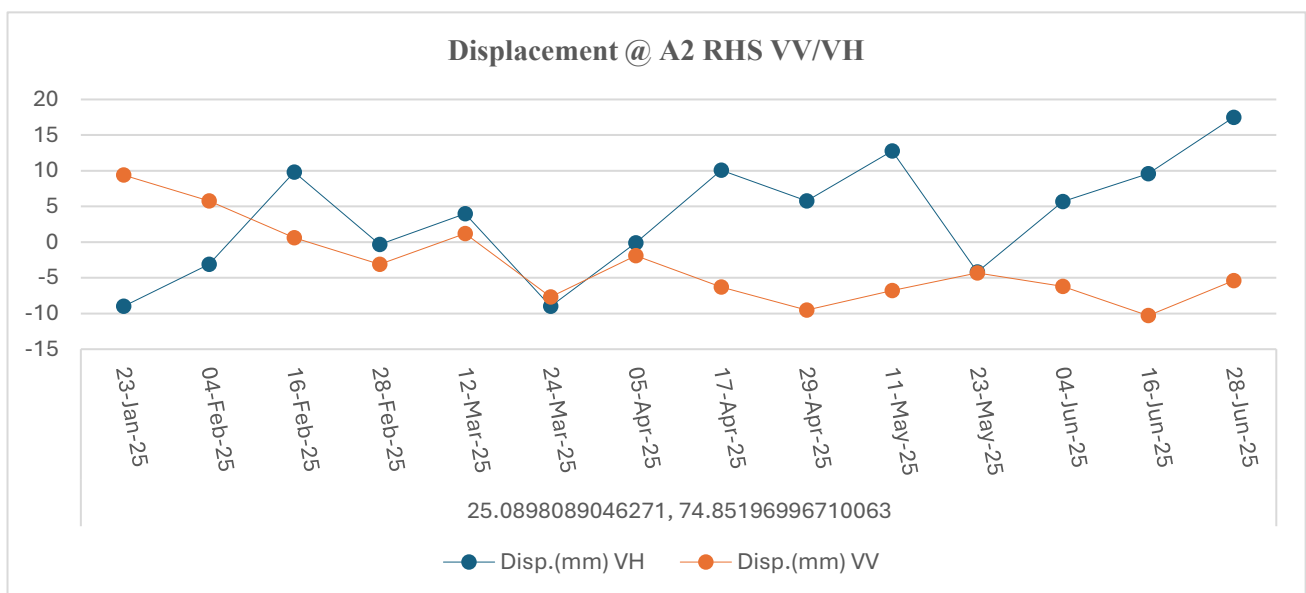
Critical Events & Anomalies:

- **January-February 2025:** Both VH and VV remain low and stable (VH around -2 to $+1$ mm; VV around $+2$ to $+4$ mm), indicating steady structural behaviour.
- **June 2025:** VH slightly reduces but stays positive ($+8$ – $+11$ mm), while VV stabilizes around $+9$ – $+11$ mm, indicating post-peak stabilization.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (25.0898089046271, 74.85196996710063)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical (VV) displacement** VV displays a predominantly negative trend, ranging roughly from +9 mm to -11 mm.
- **Horizontal (VH) displacement** exhibits moderate variability with an overall positive tendency, ranging approximately from -8 mm to +17 mm.

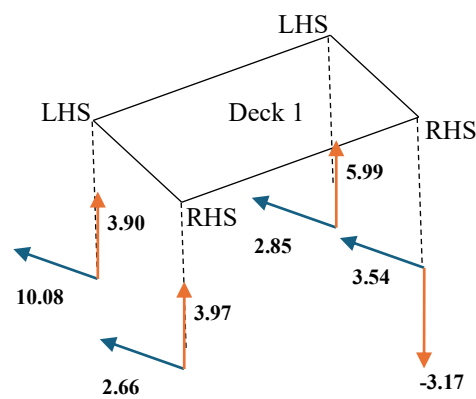
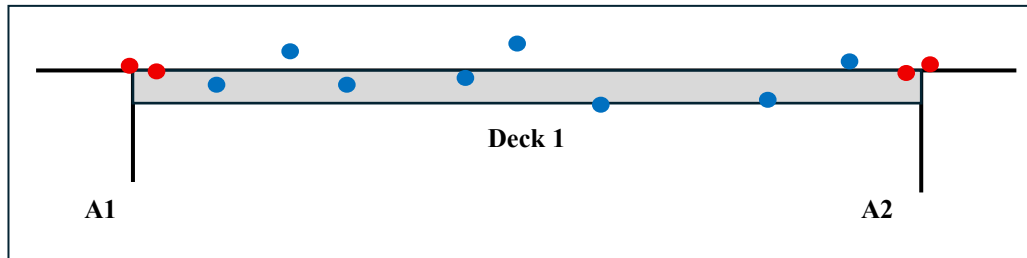
Critical Events & Anomalies:

- **January-February 2025:** VH remains negative to mildly positive while VV decreases from about +9 mm to near 0 mm, suggesting early-stage adjustment.
- **May 2025:** VH drops briefly to near -5 mm, coinciding with VV around -4 mm, indicating a temporary lateral reversal.
- **June 2025:** VH increases steadily to its maximum (+17 mm), while VV remains consistently negative (-6 to -11 mm), suggesting lateral movement dominance with ongoing vertical settlement.

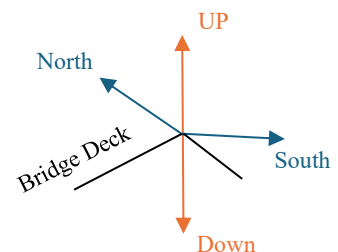
ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Bridge Deck Slab Analysis



Vertical Displacement Behaviour:

- The bridge deck slab exhibits predominantly uplift-dominated vertical behaviour, with upward movements observed at most support locations. The central lower region records a vertical uplift of approximately +3.97 mm, suggesting limited vertical differential between slab and support levels.

Horizontal Displacement Behaviour:

- Horizontal movements are moderate and directionally consistent, with a clear lateral bias. The central lower region shows a smaller horizontal displacement of approximately +2.66 mm, suggesting restrained lateral response at mid-span support.

CONCLUSION:

- Overall, the bridge deck slab shows stable structural behaviour, with vertical and horizontal displacements remaining within acceptable operational limits. The observed uplift-dominated response is consistent with thermal expansion or transient loading effects, rather than long-term settlement